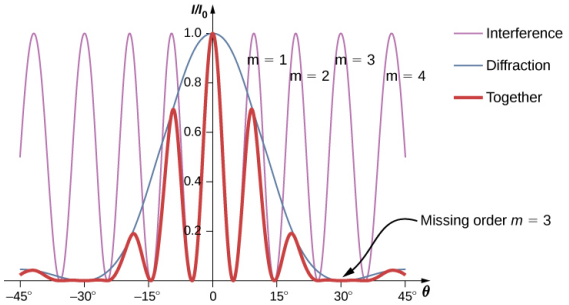
1. The figure below shows that the intensity of the fringe for m = 3 is zero. Determine the intensity of the fringe at the m=1 interference maxima relative to I_0. You can see from the figure that it is around 0.7



2. Two slits of width 2 μm, each in an opaque material are separated by a center-to-center distance of 6 μm. A monochormatic light of wavelength 450 nm is incident on the double slit. A combined diffraction plus interference pattern is found on the screen.

A. How many interference peaks ("fringes") will be obseved in the diffraction central maximum?

B. How many interference peaks will be observed in the diffraction central maximum if the slit width is doubled while keeping everything else constant.

C. How many interference peaks will be observed in the diffraction central maximum if the slits are separated by double the amount while keeping everything else constant?

D. What will happen if 680 nm light is used instead of 450 nm light.

E. What is the intensity of the peak of the first secondary diffraction lobe compared to the peak diffraction central max?

3. Use GeoGebra to plot the intensity of a double slit diffraction experiment versus angle. Create sliders for wavelength λ, slit width a, and slit separation d. Plot the intensity function:

$$I(\theta) = I_0 \ \mathrm{sinc}\,(\beta)^2 \cos^2(\alpha)$$

where $\beta = \frac{\pi a \sin\theta}{\lambda}$ and $\alpha = \frac{\pi d \sin\theta}{\lambda}$.

A. Manipulate (a) wavelength, (b) slit width, and (c) slit separation. What effect does each parameter have on the diffraction pattern? Which parameter controls the envelope? Which controls the fringes?

B. Use GeoGebra's `Max[]` function or the extremum tool to find the peak intensity of the first secondary diffraction lobe (the first side maximum of the envelope). How does this compare to the central maximum?

4. A diffraction grating has 2000 lines per centimeter. At what angle is the second order maximum for 520-nm green light?

5. The 305-m diameter Arecibo radio telescope detects radio waves with 21-cm wavelength.

A. What is the angular separation between two barely observable objects for this telescope?

B. What physical separation does this translate to for an object in the Andromeda Galaxy 2 Mly from us?

6. The structure of NaCl crystals packs atoms about 0.541 nm apart. What is the minimum angle, measured from one of the planes of atoms, where x-ray diffraction can be observed to occur if the crystal is illuminated by x-rays of wavelength 0.085 nm?